

STUDY AND EVALUATION SCHEME

Of

Diploma (Engineering)- 3Years

Diploma (Engineering)- Civil Engineering II Year

Semester -IV



**BHABHA UNIVERSITY,
BHOPAL (M.P.)**



BHABHA UNIVERSITY, BHOPAL

CREDIT BASED GRADING SYSTEM

BRANCH: CIVIL ENGG

Implemented from Session: July 2018

SUBJECT CODE	SUBJECT NAME	THEORY COMPONENT								PRACTICAL COMPONENT						TOTAL CREDIT	GRAND TOTAL OF MARKS
		LECTURES							THEORY CREDIT	PRACTICAL Hrs. Per Week	CONTI NUOUS EVALU ATION	END OF THE TERM/ SEMESTER EVALUATION			PRACTCAL CREDIT		
			TERM WORK QUIZ, ASSIG NMENT	MID TERM TEST (TWO)		THEORY PAPER					LAB. WORK QUIZ, ASSIG NMENT	PRACTICAL / ORAL EXAMINATION (VIVA)					
				I	II	NO.	MARKS	DURATION (Hrs)				NO.	MARKS	DURATION (Hrs.)			
4001	Advance Surveying	03	10	10	10	01	70	3	04	04	20	1	30	3	02	06	150
4002	Soil Mechanics	04	10	10	10	01	70	3	04	02	20	1	30	3	02	06	150
4003	Mechanics of Structure	06	10	10	10	01	70	3	04	-	-	-	-	-	-	04	150
4004	Transportation Engg-I	04	10	10	10	01	70	3	04	02	20	1	30	3	02	06	150
4005 / 4006	Optional [any one] Entrepreneurship\ Marketing management	06	10	10	10	01	70	3	04	-	-	-	-	-	-	04	100
4007	Computer Aided Drawing									03	40	1	60	3	02	02	100
4008	Professional activities									02							
	TOTAL	23	50	50	50	05	350		20	13	100		150		08	28	800

Theory Credits : 20	Theory Marks : 350
Practical Credits : 10	Practical Marks : 150
Total Credits : 30	Quiz, Mid Term, Lab Work : 250
	Total Credits : 700

BHABHA UNIVERSITY, BHOPAL

Subject – Advance Surveying

Sub Code –CE 4001

UNIT-I

PLANE TABLE SURVEY

Principles of plane table survey. Accessories required. Setting out of plane table, Leveling, Centering and orientation. Methods of plane table surveying – Radiation, Intersection, and Traversing. Merits and Demerits of planetable. Surveying situations where plane table survey is used. Use of Telescopic Alidade.

UNIT-II

THEODOLITE SURVEY

Components of Transit Theodolite and Their functions. Technical terms used. Temporary adjustments of Transit Theodolite. Swinging the telescope, Transiting, Changing the face. Measurement of Horizontal angle, method of Repetition, errors eliminated by method of repetition. Measurement of Deflection angle. Measurement of Vertical angle. Measurement of magnetic bearing of a line by Theodolite. Prolonging a Straight line. Sources of errors in Theodolite Surveying. Permanent adjustment of transit Theodolite (only relationship of different axes of Theodolite.) Traversing with Theodolite – Method of included angles, locating details, checks in closed traverse, Calculation of bearings from angles. Traverse Computation - Latitude, Departure Consecutive Co-ordinates error of Closure, Distribution of an angular error, balancing the traverse by Bowditch rule and Transit Rule, Gale's traverse table. simple problems on above topic.

UNIT-III

TACHEOMETRIC SURVEY

Principle of Tacheometry. Essential requirements of Tacheometer. Use of Theodolite as a Tacheometer with staff held in vertical and fixed hair method (No

derivation).Determination of tacheometric constants, simple numerical problems on above topics.

UNIT-IV

CURVES

Types of curves used in road and railway alignments. Notations of simple circular curve. Designation of curve by radius and degree of curves. Method of Setting out curve by offset from Long chord method and Rankine's method of deflection. angles. Simple Numerical problems on above topics.

UNIT-V

ADVANCED SURVEY EQUIPMENTS

Construction and use of one second Micro Optic Theodolite, Electronic Digital Theodolite. Features of Electronic Theodolite Principle of E.D.M, Components of E.D.M and their functions, use of E.D.M. Total station

AERIAL SURVEY AND REMOTE SENSING

Aerial Survey

Introductions, definition, Aerial photograph. Remote Sensing – Introduction, Electro-Magnetic Energy , Remote sensing system- Passive system , Active system. Applications – mineral, land use / Land cover, Natural Hazards and Environmental engineering system.

LIST OF EXPERIMENTS

- Using accessories carry out temporary adjustments of plane table.
Locating details by method of Radiation.
- Locating details with plane table by method of intersection.
- Understanding the components of Theodolite and their functions, reading the vernier and temporary adjustments of theodolite.
- Measurement of Horizontal angle by transit theodolite.

- Measurement of Horizontal angle by method of Repetition.
- Measurement of vertical angles by theodolite.
- Measurement of Magnetic bearing of a line using theodolite.
- Measurement of deflection angle by taking open traverse of 4 –5 sides.
- To find Reduced levels and horizontal distances using theodolite as a Tacheometer.
- To find constants of a given Tacheometer.
- Study and use of 1 second Micro Optic Theodolite for measurement of Horizontal and Vertical angles.
- Study of E.D.M. For knowing its components.

LIST OF REFERENCE BOOKS

S. No.	Title	Author	Publisher
1	Surveying and Levelling,	N N Basak,	Tata Mc Graw-Hill
2.	Surveying and Levelling Part I and II,	T .P. Kanetkar & S. V, Kulkarni,	Pune Vidhyarthi Griha Prakashan.
3	Surveying and Levelling Vol. I and II	Dr. B. C. Punamiya	Laxmi Publication
4	Text book of Surveying,	S.K.Husain, M.S. Nagaraj	S. Chand and company
5	Surveying and Levelling Vol. I and II,	S. K. Duggal,	Tata Mc Graw-Hill
4	Plane Surveying,	A.M.Chandra,	New Age International Publishers

BHABHA UNIVERSITY, BHOPAL

SUBJECT – SOIL MECHANICSSUB

CODE –CE 4002

UNIT-I

PHYSICAL PROPERTIES OF SOIL

Soil as a three phase system. Water content, Determination of water content by oven drying method as per IS code. Void ratio, porosity and degree of saturation, density index. Unit weight of soil mass – bulk unit weight, dry unit weight, unit weight of solids, saturated unit weight, submerged unit weight.

Determination of bulk unit weight and dry unit weight by core cutter method and sand replacement method as per IS code. Specific gravity, determination of specific gravity by pycnometer

UNIT-II

CLASSIFICATION OF SOIL

Field identification tests of fine grained soil, IS. classification chart. Consistency of soil, stages of consistency, Atterberg's. limits of consistency viz. Liquid limit, plastic limit and shrinkage limit, plasticity index. Determination of liquid limit, plastic limit and shrinkage limit as per IS code. Classification of fine grained soil by using plasticity chart. Seive analysis of soil and sedimentation of soil, log, scale of particle size. Stokes law, Consistency limit diagram. Particle size distribution, mechanical sieve analysis as per. IS code particle size distribution curve, effective diameter of soil, Uniformity coefficient and coefficient of curvature, well graded and uniformly graded soils. Particle size classification of soils & IS classification of soil.

PERMEABILITY OF SOIL & SEEPAGE ANALYSIS :

Definition of permeability. Laminar and turbulent flow. Importance of permeability. Darcy's law of permeability, coefficient of permeability, typical values of coefficient of permeability for different soil. Factors affecting permeability. Determination of

coefficient of permeability by constant head and falling head permeability tests, simple problems to determine coefficient of permeability. Seepage through earthen structures, seepage velocity, seepage pressure, phreatic line, flow lines and equipotential lines. Flow net, characteristics of flow net, application of flow net (no numerical problems)

UNIT-III

SHEAR STRENGTH OF SOIL

Shear failure of soil, field situation of shear failure. Concept of shear strength of soil. Components of shearing resistance of soil – cohesion, internal friction. Mohr-coulomb failure theory (Coulomb's Law), Strength envelope, strength Equation. Purely cohesive and cohesion less soils. Laboratory determination of shear strength of soil – Direct shear test, Box shear test and tri-axial test Unconfined compression test & vane shear test, plotting strength envelope, determining shear strength parameters of soil.

BEARING CAPACITY OF SOILS AND EARTH PRESSURE

Concept of bearing capacity, ultimate bearing capacity, safe bearing capacity and allowable bearing pressure. Terzaghi's analysis and assumptions made. Effect of water table on bearing capacity. Field methods for determination of bearing capacity – Plate load test and standard penetration test. Test procedures as Per IS:1888 & IS:2131. Typical values of bearing capacity from building code IS:1904. Definition of active earth pressure and passive earth pressure, structures subjected to earth pressure in the field. Earth pressure, effective pressure. Neutral pressure, and total pressure Magnitude of earth pressure. Rankine's theory, Assumptions made in the Rankine's theory. Earth retaining structures. Earth pressure on earth retaining structures. Bearing capacity of soil during earthquake.

UNIT-IV

COMPACTION OF SOIL & STABILIZATION:

Concept of compaction, purpose of compaction field situations where compaction is required. Standard proctor test – test procedure as per IS code, Compaction curve, optimum moisture content, maximum dry density, Zero air voids line. Modified proctor test. Factors affecting compaction. Field methods of compaction – rolling, ramming & vibration and Suitability of various compaction equipments. California bearing ratio, CBR test, significance of CBR value. Difference between compaction and consolidation. Concept of soil stabilization, necessity of soil stabilization. Different methods of soil stabilization – Mechanical soil stabilization, lime stabilization, cement stabilization, bitumen stabilization, fly-ash stabilization

UNIT-V

SITE INVESTIGATION AND SUB SOIL EXPLORATION

Necessity of site investigation & sub-soil exploration. Types of exploration – general, detailed. Method of site exploration open excavation & boring. Criteria for deciding the location and number of test pits and bores. Trial pits types of Augers. Auger boring, wash boring and percussion drilling. Disturbed & undisturbed soil samples for lab testing. Field identification of soil – dry strength test, dilatancy test & toughness test. Empirical correlation between soil properties and SPT values. Record of boring Bore hole log.

LIST OF EXPERIMENTS

- Determination of water content of given soil sample by oven drying method as per IS Code.
- Determination of bulk unit weight dry unit weight of soil in field by core cutter method as per IS Code.

- Determination of bulk unit weight dry unit weight of soil in field by sand replacement method as per IS Code.
- Determination of Liquid limit & Plastic limit of given soil sample as per IS Code.
- Determination of grain size distribution of given soil sample by mechanical sieve analysis as per IS Code
- Determination of coefficient of permeability by constant head test.
- Determination of coefficient of permeability by falling head test Practical (Live demo or Prerecorded demo)
- Determination of shear strength of soil using direct shear test.
- Determination of shear strength of soil using Laboratory Vane shear test.
- Determination of MDD & OMC by standard proctor test on given soil sample as per IS Code.
- Determination of CBR value of given soil sample.
- Determination of shear strength of soil using tri-axial shear test.

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Soil Mechanics & Foundation Engineering,	Dr. B. C. Punmia,	Standard Book house, New Delhi.
2.	Soil Mechanics & Foundation Engineering,	V.N.S. Murthi	Tata McGraw Hill , New Delhi.
3	Soil Mechanics,	B. J. Kasmalkar	Pune VidhyartiGriha, Pune
4	Geo-technical Engineering,	Gulhati& Dutta	Tata McGraw Hill , New Delhi

BHABHA UNIVERSITY, BHOPAL

Subject –Mechanics of Structure

SubCode –CE 4003

UNIT-I

STRESS & STRAIN

Definition of rigid body, plastic body, mechanical properties of metal such as elasticity & elastic limit .Definition of stress, strain, modulus of elasticity, S.I. Unit. Classification of stress, strain, Sign convention. Stress, strain curve for mild steel and HYSD bar ,yield stress/ proof stress, Ultimate stress, breaking stress and percentage elongation. Deformation of body due to axial load. Deformation of a Body subjected to axial forces. Deformation of body of stepped c/s due to axial load, max. stress and min. stress induced. Stresses in bars of composite section & deformation. Shear stress, shear strain & modulus of rigidity, complementary shear stress, state of simple shear, punchingshear.

ELASTIC CONSTANTS & PRINCIPAL STRESSES

Definition of lateral strain, Poisson's ratio, Change in lateral dimensions. Volumetric strain due to uni-axial force and change in volume. Biaxial and tri-axial stresses and volumetric strain & change in volume. Definition of bulk modulus, volumetric strain. Relation between modulus of elasticity, modulus of rigidity and bulk modulus. Definition of principal planes & principal stresses. Principal planes & stress due to bi-axial stress system & due to state of simple shear (Analytical method only). Strain Energy : Types of loading – gradual, suddenly applied load & Impact load. Definition of strain energy, modulus of resilience and proof resilience. Comparison of stresses due to gradual load, sudden load and impact load.

UNIT-II

SHEAR FORCE AND BENDING MOMENT

Types of beams - cantilever, simply supported, fixed and continuous beams, types of loading- point load, uniformly distributed load, support reactions for determinate structures. Concept of shear force and bending moment, sign convention. Relation between bending moment, shear force and rate of loading. Shear force and bending moment diagrams for simply supported beams, overhanging beams and cantilever subjected to point loads, UDL and couples, point of contraflexure.

UNIT-III

MOMENT OF INERTIA

Concept of moment of inertia, M.I of plane areas such as rectangle, triangle, circle, semicircle and quarter circle. Parallel axis and perpendicular axis theorem, M.I of composite sections, built up sections, symmetrical and unsymmetrical sections, radius of gyration & polar moment of inertia.

UNIT-IV

STRESSES IN BEAMS

Bending Stresses in Beams: Concept of pure bending, theory of simple bending, assumptions in theory of bending, neutral axis, bending stresses and their nature, bending stress distribution diagram, moment of resistance. Application of theory of bending to symmetrical and unsymmetrical sections. Shear stresses in beams, Shear stress equation, meaning of terms in equation, shear stress distribution for rectangular, hollow rectangular, circular sections and hollow circular sections, I sections and T sections. Relation between max. shear stress and average shear stress.

UNIT-V

ANALYSIS OF TRUSSES

Definition frames, classification of frames, perfect, imperfect, redundant and deficient frame, relation between members and joints, assumption in analysis.

Method of joint, method of section and graphical method to find nature of forces.

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Strength of Materials	F. L. Singer,	Harpe Collins Publishers India, Delhi
2.	Strength of Materials,	R. S. Khurmi,	S. Chand & Company, Delhi
3	Mechanics of Structures,	S. B. Junnarkar- volume –I & II,	Charotar Publishing House, Anand.
4	Strength of Materials,	Sadhu Singh.	

BHABHA UNIVERSITY, BHOPAL

Subject – Transportation Engineering-I

Sub Code –CE 4004

UNIT I

OVERVIEW OF TRANSPORTATION ENGINEERING

Role of transportation in the development of nation. Modes of transportation system – roads, railway, airways, waterways, Importance of each mode, comparison and their relative merits and demerits. Necessity & importance of Cross drainage works for roads & railways.

UNIT II

RAILWAY ENGINEERING

Alignment and Gauges, Classification of Indian Railways, zones of Indian Railway. Alignment- Factors governing rail alignment. Rail Gauges – types, factors affecting selection of gauge. Rail track cross sections – standard cross section of BG & M.G Single & double line in cutting and embankment. Permanent ways.

UNIT III

IDEAL REQUIREMENT, COMPONENT PARTS

Rails –function & its types. Rail Joints – requirements, types, Creep of rail- causes & prevention of creep. Sleepers – functions & Requirement, types – wooden, metal, concrete sleepers & their suitability, sleeper density. Ballast – function & different types with their properties, relative merits & demerits. Rail fixtures & fastenings – fish plate, bearing plates, spikes, bolts, keys, anchors & anti creepers. Railway Track Geometrics. Coning of wheels, tilting of rails, Gradient & its types, Super elevation, limits of Super elevation on curves, Cant deficiency, negative

cant, grade compensation on curves. Branching of Tracks. Definition of point & crossing, a simple split switch turnout consisting of points and crossing lines. Sketch showing different components, their functions & working. Line sketches of track junctions-crossovers, scissor cross over, diamond crossing, triangle. Inspection of points and crossings. Station and Yards : Site selection for railway stations, Requirements of railway station, Types of stations (way side, crossing, junction & terminal) Station yards , types of station yard, Passenger yards, Goods yard Locomotive yard, its requirements, water column ,Marshalling, yard, its types. Track Maintenance - Necessity, types, Tools require and their function, organization, duties of permanent way inspector, gang mate, key man

UNIT IV

BRIDGE ENGINEERING

Site selection and investigation Factors affecting selection of site of a bridge. Bridge alignment Collection of design data Classification of bridges according to function, material, span, size, alignment, position of HFL. Component parts of bridge. Plan & sectional elevation of bridge showing component parts of substructure & super structure. Different terminology such as effective span, clear span, economical span, waterway, afflux, scour, HFL, freeboard, etc. Foundation – function, types Piers-function, requirements, types. Abutment – function, types, Wing walls – functions and types. Bearing – functions, types of bearing for RCC & steel bridges. Approaches – in cutting and embankment. Bridge flooring- open and solid floors. Permanent and Temporary Bridges- Permanent Bridges - Sketches & description in brief of culverts, causeways, masonry, arch, steel, movable steel bridges, RCC girder bridge, pre-stressed girder bridge, cantilever, suspension bridge. Temporary Bridges- timber, flying, floating bridges Inspection & Maintenance Of Bridge - Inspection of bridges, Maintenance of bridges & types, routine & special maintenance.

UNIT V

TUNNEL ENGINEERING

Definition, necessity, advantages, disadvantages. Classification of tunnels. Shape and Size of tunnels. Tunnel Cross sections for highway and railways. Tunnel investigations and surveying –Tunnel surveying locating center line on ground, transferring center line inside the tunnel. Shaft - its purpose & construction. Methods of tunneling in Soft rock - needle beam method, fore-poling method. line plate method, shield method. Methods of tunnelling in Hard rock – Full face heading method, Heading and bench method, drift method. Precautions in construction of tunnels. Drilling equipments-drills and drills carrying equipments. Types of explosives used in tunnelling. Tunnel lining and ventilation.

TRANSPORTATION ENGINEERING – I

Visits & Report Generation :Student will have to prepare at least 08 reports on visits to different maintenance and operations related to railway tracks during visits. The write-ups for the reports should include following information :

- (i) Objects of maintenance operations.
- (ii) Materials required.
- (iii) Tools and Equipments needed.
- (iv) Maintenance procedure.
- (v) Precautions to be taken during maintenance operations.
- (vi) Remedial measures and quality control to reduce the maintenance requirements.

TOPICS FOR VISITS & REPORTS

1. Throughpacking
2. Shovelpacking
3. Track maintenance

4. System aticoverhauling
5. Lifting of track
6. Lowering of track
7. Counteraction, measurement and adjustment of creep
8. Organization, Tools and equipments for maintenance.
9. Maintenance of points and crossings
10. Maintenance of level crossing.
11. Maintenance of proper Drainage Visit for cross drainage works for roadways and railways

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Railway Engineering,	S.C. Saxena	Dhanpatrai& sons
2.	Railway Track	K.R. Antia,	The New Book Co. Pvt. Ltd Mumbai
3	Principles of Railway Engineering	S.C. Rangwala,	Charotar Publication.
4	Principles and Practice of Bridge Engineering,	S.P. Bindra	Dhanpatrai& sons.
5	A Text Book of Transportation Engineering	N.L.Arora and S.P.Luthra,	IPH New Delhi.

BHABHA UNIVERSITY, BHOPAL

Subject – Entrepreneurship

Sub Code –CE 4005

UNIT -I

INTRODUCTION TO ENTREPRENEURSHIP

- Definition of Entrepreneur /Entrepreneur
- Difference between Entrepreneurship /Entrepreneurship
- Need forEntrepreneurship
- qualities of successfulentrepreneur
- Myths about Entrepreneurship
- Classification of entrepreneurs on the basis ofdifferent criteria
- Reasons for the failure ofentrepreneurs

UNIT –II

INDUSTRIES AND BUSINESS ORGANIZATIONS

- Concept of Industry orEnterprise
- Classification ofIndustries
 - (a) On the basis of capitalinvestment
 - Tiny (Micro)Industry
 - SmallScale
 - MediumScale
 - LargeScale
 - (b) Others
 - RuralIndustry
 - CottageIndustry
 - (c) Forms of BusinessOrganization
 - Proprietorship
 - Board &Co-operative
 - Partnership
 - PublicLtd.

- Private Ltd.
- IT Sector
- Government Co-operative / Undertakings

(d) Tiny small scale Industry

- Definition
- Its significance in National Development.
- Govt. policies for SSI promotions
- Sector / Product for SSI.

INSTITUTIONAL ASSISTANCE

(a) Types of Institutional assistance

- Infra - structural assistance
- Technical Assistance
- Financial assistance
- Marketing Assistance

(b) Information / guidance & Training

- | | |
|----------|--------|
| - SISI | - ASK |
| - MP CON | - CSIR |
| - CED-MA | - NRDC |

(c) Infrastructure

- | | |
|-------|------------|
| - D/C | - AVN/AKVN |
|-------|------------|

(e) Finance

- | | |
|----------|--------------|
| - SIDBI | - KVIB MPFC |
| - NABARD | - MPWDC NSIC |

M.P.A.V.V.N.

(d) Marketing

- MP- AGRO
- NSIC
- PM.LUN
- EXPORT CORPORATION

- KVIP
- MPHSVN MPLDC

(e) Quality Control

- BIS -FPO -MPLUN F.D.A.
- AG. MKT. Board

UNIT –IV

INCENTIVES / CONCESSION / FACILITIES AVAILABLE

- Seed money
- Incentive / subsidies

Others (Phones, Landsetc)

UNIT –V

PLANNING OF AN INDUSTRIAL UNIT (SSI)

- Pre- PlanningStage
 - Scanning theenvironment
 - Marketsurvey
 - Seekinginformation
 - product / projectselection
- Implementation Stage
 - PPRPreparation
 - DICregistration

PROJECT WORK/ASSIGNMENT

1. To prepare chart to showing various factors affectingentrepreneurship.
2. To collect details related to various schemes run by the Govt. for Self-Employment and Entrepreneurship.
3. To identify and select a project and conduct Market-Surveythereof.
4. To collect various formats used in industries & departments/institutions working inthe field

of entrepreneurship.

5. Visit few small scale industries situated in city, nearby industrial area.
6. Discuss the problems related to SSI (Small Scale Industries) with an entrepreneur.
7. Collect information about market rates quality and quantity of goods for their choice.
8. Develop logical and analytical approach to purchase the raw material / finished goods.
9. To prepare case study of successful entrepreneurs. Preparation of Project report for the industry/ Business they are willing to start

LIST OF REFERENCE

S. No.	Title	Author	Publisher
1	Entrepreneurial Development Vol. I,II,III	By Vasant desai	Himalaya Publication
2	CEDMAP	(Center of Entrepreneurial development Madhya Pradesh)	